

Configuration properties

An agent needs some parameters to be set to properly work. In this section, you will find all of them, and we will explain you how to load them in the agent.

An agent Property has a name, a type and a default value (for when the property is not defined by the agent developer).

Property names follow flattened dot notation from the YAML hierarchy, e.g.
platforms.websocket.host.

Loading properties

You can define your agent properties in 2 different ways:

Using a configuration `.yaml` file: It is a file containing all the agent properties. Let's see an example

config.yaml file:

agent:

check_transitions_delay: 5

nlp:

language: en

region: US

timezone: Europe/Madrid

pre_processing: True

intent_threshold: 0.4

huggingface:

token: YOUR-TOKEN

openai:

api_key: YOUR-API-KEY

replicate:

api_key: YOUR-API-KEY

platforms:

websocket:

host: localhost

port: 8765

origins:

- "https://example.com"

- "https://app.example.com"

streamlit:

host: localhost

port: 5000

chat:

size: 16

font: sans

line_spacing: 1.5

alignment: left

```
    color: inherit
    contrast: medium
telegram:
  token: YOUR-BOT-TOKEN
github:
  personal_token: YOUR-PERSONAL-TOKEN
  webhook_token: YOUR-WEBHOOK-TOKEN
  webhook_port: 8901
gitlab:
  personal_token: YOUR-PERSONAL-TOKEN
  webhook_token: YOUR-WEBHOOK-TOKEN
  webhook_port: 8901
a2a:
  port: 8000
```

```
db:
  monitoring:
    enabled: True
    dialect: postgresql
    host: YOUR-DB-HOST
    port: 5432
    database: YOUR-DB-NAME
    username: YOUR-DB-USERNAME
    password: YOUR-DB-PASSWORD
  streamlit:
    enabled: True
    dialect: postgresql
    host: YOUR-DB-HOST
    port: 5432
    database: YOUR-DB-NAME
    username: YOUR-DB-USERNAME
    password: YOUR-DB-PASSWORD
  sql:
    - db1:
        dialect: sqlite
        database: path/to/local.db
    - db2:
        dialect: postgresql
        host: YOUR-DB-HOST
        port: 5432
        database: YOUR-DB-NAME
        username: YOUR-DB-USERNAME
        password: YOUR-DB-PASSWORD
```

Now you have to load the file into the agent:

```
agent = Agent('example_agent')
agent.load_properties('config.yaml')
```

Setting individual properties: You can also set (and get) properties individually from the agent code.

```
from baf.nlp import NLP_LANGUAGE
...
agent = Agent('example_agent')
agent.set_property(NLP_LANGUAGE, 'es')
...
language = agent.get_property(NLP_LANGUAGE)
Note
```

When you try to get a property that has not been previously set, it will return its default value.

You can also create your own properties:

```
from baf.core.property import Property
...
FACEBOOK_PROFILE = Property('facebook.profile', str, 'https://www.facebook.com/foo')
...
agent.set_property(FACEBOOK_PROFILE, 'https://www.facebook.com/john_doe')
Next, let's see all built-in properties, grouped by domain.
```

Agent

Definition of the agent properties within the agent section:

`baf.CHECK_TRANSITIONS_DELAY = <baf.core.property.Property object>`

An agent evaluates periodically all the transitions from the current state to check if someone is satisfied.

This property sets the delay between each transitions evaluation, in seconds.

name: `agent.check_transitions.delay`

type: `float`

default value: `1`

NLP

Definition of the agent properties within the nlp (Natural Language Processing) section

`baf.nlp.HF_TOKEN = <baf.core.property.Property object>`

The HuggingFace (Inference) API key, necessary to use a HuggingFace Inference API LLM.

name: `nlp.huggingface.token`

type: `str`

default value: `None`

baf.nlp.NLP_INTENT_THRESHOLD = <baf.core.property.Property object>

The threshold for the Intent Classification problem. If none of its predictions have a score greater than the threshold, it will be considered that no intent was detected with enough confidence (and therefore, moving to a fallback scenario).

name: nlp.intent_threshold

type: float

default value: 0.4

baf.nlp.NLP_LANGUAGE = <baf.core.property.Property object>

The agent language. This is the expected language the users will talk to the agent. Using another language may affect the quality of some NLP processes.

The list of available languages can be found at [snowballstemmer](#). Note that luxembourgish (lb) is also partially supported, as the language can be chosen, yet the stemmer is still a work in progress.

Languages must be written in ISO 639-1 format (e.g., 'en' for English)

name: nlp.language

type: str

default value: en

baf.nlp.NLP_PRE_PROCESSING = <baf.core.property.Property object>

Whether to use text pre-processing or not. Stemming is the process of reducing inflected (or sometimes derived) words to their word stem, base or root form.

Currently, only SimpleIntentClassifierTorch, SimpleIntentClassifierTF and SimpleNER use this property. If LLMIntentClassifier is used, this property is ignored.

For example 'games' and 'gaming' are stemmed to 'game'.

It can improve the NLP process by generalizing user inputs.

name: nlp.pre_processing

type: bool

default value: True

baf.nlp.NLP_REGION = <baf.core.property.Property object>

The language region. If specified, it can improve some NLP process You can find a list of regions [here](#).

name: nlp.region

type: str

default value: US

baf.nlp.NLP_TIMEZONE = <baf.core.property.Property object>

The timezone. It is used for datetime-related tasks, e.g., to get the current datetime. A list of timezones can be found [here](#).

name: nlp.timezone

type: str

default value: Europe/Madrid

baf.nlp.OPENAI_API_KEY = <baf.core.property.Property object>

The OpenAI API key, necessary to use an OpenAI LLM.

name: nlp.openai.api_key

type: str

default value: None

baf.nlp.REPLICATE_API_KEY = <baf.core.property.Property object>

The Replicate API key, necessary to use a Replicate LLM.

name: nlp.replicate.api_key

type: str

default value: None

WebSocket Platform

baf.platforms.websocket.STREAMLIT_CHAT_DEFAULT_ALIGNMENT =

<baf.core.property.Property object>

Default chat text alignment used by Streamlit chat UI.

name: platforms.websocket.streamlit.chat.alignment

type: str

default value: left

baf.platforms.websocket.STREAMLIT_CHAT_DEFAULT_COLOR =

<baf.core.property.Property object>

Default chat text color used by Streamlit chat UI.

name: platforms.websocket.streamlit.chat.color

type: str

default value: inherit

baf.platforms.websocket.STREAMLIT_CHAT_DEFAULT_CONTRAST =
<baf.core.property.Property object>

Default chat contrast level used by Streamlit chat UI.

name: platforms.websocket.streamlit.chat.contrast

type: str

default value: medium

baf.platforms.websocket.STREAMLIT_CHAT_DEFAULT_FONT =
<baf.core.property.Property object>

Default chat font family key used by Streamlit chat UI.

name: platforms.websocket.streamlit.chat.font

type: str

default value: sans

baf.platforms.websocket.STREAMLIT_CHAT_DEFAULT_LINE_SPACING =
<baf.core.property.Property object>

Default chat line spacing used by Streamlit chat UI.

name: platforms.websocket.streamlit.chat.line_spacing

type: float

default value: 1.5

baf.platforms.websocket.STREAMLIT_CHAT_DEFAULT_SIZE = <baf.core.property.Property object>

Default chat font size used by Streamlit chat UI when no profile-specific style is applied.

name: platforms.websocket.streamlit.chat.size

type: int

default value: 16

baf.platforms.websocket.STREAMLIT_HOST = <baf.core.property.Property object>
The Streamlit UI host address. If you are using our default UI, you must define its address where you can access and interact with the agent.

name: platforms.websocket.streamlit.host

type: str

default value: localhost

baf.platforms.websocket.STREAMLIT_PORT = <baf.core.property.Property object>
The Streamlit UI address port. The Streamlit UI address is composed by a host name and a port

name: platforms.websocket.streamlit.port

type: int

default value: 5000

baf.platforms.websocket.WEBSOCKET_HOST = <baf.core.property.Property object>
The WebSocket host address. An agent has a WebSocket server that has to establish connection with a WebSocket client.

name: platforms.websocket.host

type: str

default value: localhost

baf.platforms.websocket.WEBSOCKET_MAX_SIZE = <baf.core.property.Property object>
WebSocket's maximum size of incoming messages, in bytes. None disables the limit.

name: platforms.websocket.max_size

type: int

default value: None

baf.platforms.websocket.WEBSOCKET_ORIGINS = <baf.core.property.Property object>
List of allowed origins for WebSocket connections. When set, only connections from these origins will be accepted. When None (default), connections from any origin are allowed.

Example in config.yaml:

```
platforms:
  websocket:
    origins:
```

- "https://editor.besser-pearl.org"
- "https://localhost"

name: platforms.websocket.origins

type: list

default value: None

baf.platforms.websocket.WEBSOCKET_PORT = <baf.core.property.Property object>

The WebSocket address port. The WebSocket address is composed by a host name and a port

name: platforms.websocket.port

type: int

default value: 8765

Telegram Platform

Definition of the agent properties within the telegram_platform section:

baf.platforms.telegram.TELEGRAM_TOKEN = <baf.core.property.Property object>

The Telegram Bot token. Used to connect to the Telegram Bot

type: str

default value: None

GitHub Platform

Definition of the agent properties within the github_platform section:

baf.platforms.github.GITHUB_PERSONAL_TOKEN = <baf.core.property.Property object>

The Personal Access Token used to connect to the GitHub API

name: github.personal_token

type: str

default value: None

baf.platforms.github.GITHUB_WEBHOOK_PORT = <baf.core.property.Property object>

The server local port. This port should be exposed or proxied to make it visible by GitHub

name: github.webhook_port

type: int

default value: 8901

baf.platforms.github.GITHUB_WEBHOOK_TOKEN = <baf.core.property.Property object>
The secret token defined at the webhook creation

name: github.webhook_token

type: str

default value: None

GitLab Platform

Definition of the agent properties within the gitlab_platform section:

baf.platforms.gitlab.GITLAB_PERSONAL_TOKEN = <baf.core.property.Property object>
The Personal Access Token used to connect to the GitLab API

name: gitlab.personal_token

type: str

default value: None

baf.platforms.gitlab.GITLAB_WEBHOOK_PORT = <baf.core.property.Property object>
The server local port. This port should be exposed or proxied to make it visible by GitLab

name: gitlab.webhook_port

type: int

default value: 8901

baf.platforms.gitlab.GITLAB_WEBHOOK_TOKEN = <baf.core.property.Property object>
The secret token defined at the webhook creation

name: gitlab.webhook_token

type: str

default value: None

A2A Platform

Definition of the agent properties within the A2A_platform section:

baf.platforms.a2a.A2A_WEBSOCKET_PORT = <baf.core.property.Property object>
The server local port. This port should be exposed or proxied to make it visible by other Agents

name: a2a.port

type: int

default value: 8000

Database

Definition of the agent properties within the db (Database) section

baf.db.DB_MONITORING = <baf.core.property.Property object>

Whether to use the monitoring database or not. If true, all the DB_MONITORING_* properties must be properly set.

name: db.monitoring.enabled

type: bool

default value: False

baf.db.DB_MONITORING_DATABASE = <baf.core.property.Property object>

The database name.

name: db.monitoring.database

type: str

default value: None

baf.db.DB_MONITORING_DIALECT = <baf.core.property.Property object>

The database dialect (e.g., postgresql).

name: db.monitoring.dialect

type: str

default value: None

baf.db.DB_MONITORING_HOST = <baf.core.property.Property object>

The database host address (e.g., localhost).

name: db.monitoring.host

type: str

default value: None

baf.db.DB_MONITORING_PASSWORD = <baf.core.property.Property object>

The database password.

name: db.monitoring.password

type: str

default value: None

baf.db.DB_MONITORING_PORT = <baf.core.property.Property object>
The database port (e.g., 5432).

name: db.monitoring.port

type: int

default value: None

baf.db.DB_MONITORING_USERNAME = <baf.core.property.Property object>
The database username.

name: db.monitoring.username

type: str

default value: None

Streamlit Database

Definition of the agent properties within the websocket_platform section:

baf.platforms.websocket.streamlit_ui.DB_STREAMLIT = <baf.core.property.Property object>
Enable Streamlit user database persistence.

name: db.streamlit.enabled type: bool default value: False

baf.platforms.websocket.streamlit_ui.DB_STREAMLIT_DATABASE =
<baf.core.property.Property object>
Database name for Streamlit user database.

name: db.streamlit.database type: str default value: None

baf.platforms.websocket.streamlit_ui.DB_STREAMLIT_DIALECT =
<baf.core.property.Property object>
Database dialect for Streamlit user database.

name: db.streamlit.dialect type: str default value: postgresql

baf.platforms.websocket.streamlit_ui.DB_STREAMLIT_HOST = <baf.core.property.Property object>
Database host for Streamlit user database.

name: db.streamlit.host type: str default value: None

baf.platforms.websocket.streamlit_ui.DB_STREAMLIT_PASSWORD =
<baf.core.property.Property object>
Database password for Streamlit user database.

name: db.streamlit.password type: str default value: None

baf.platforms.websocket.streamlit_ui.DB_STREAMLIT_PORT = <baf.core.property.Property
object>
Database port for Streamlit user database.

name: db.streamlit.port type: int default value: 5432

baf.platforms.websocket.streamlit_ui.DB_STREAMLIT_USERNAME =
<baf.core.property.Property object>
Database username for Streamlit user database.

name: db.streamlit.username type: str default value: None

Supported Languages

The following list reflects the languages supported by the BAF for which we additionally tested their correctness:

Language Name

ISO 639-1 Code

Pre Processor Source Code

Pre Processing Type

English

en

Snowball stemmer

Stemmer

Spanish

es

Snowball stemmer

Stemmer

French

fr

Snowball stemmer

Stemmer

German

de

Snowball stemmer

Stemmer

Portuguese

pt

Snowball stemmer

Stemmer

Catalan

ca

Snowball stemmer

Stemmer

Luxembourgish

lb

spellux

Lemmatizer

Snowball stemmer

The snowball stemmer supports more languages than the ones displayed in the table above. For the complete list check out <https://snowballstem.org/algorithms/>.

Luxembourgish Pre-processing

To support the luxembourgish language, we wanted to enable luxembourgish-specific text-processing. For that purpose, we opted for the best available tool, which is the

lemmatizer created by Christoph Purschke as part of “spellux - Automatic text normalization for Luxembourgish”.